

INTERWAM: INTERACTION-CENTRIC WORLD ACTION MODELS FOR PHYSICAL INTENT AND CONSEQUENCE UNDERSTANDING

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ABSTRACT

World Action Models (WAMs) jointly predict future observations and actions, offering a promising path toward generalist robot policies. Existing WAMs are predominantly trained on *successful* demonstration trajectories with vision as the primary sensory channel. We argue that this recipe yields models that imitate motion patterns without grounding in *physical interactions*: what contact is intended, whether it succeeded, and what consequences follow from a failed or partial interaction. We propose **InterWAM**, an **interaction-centric** WAM that reorganizes learning around *interaction events* rather than full task episodes. Each event is a temporally localized segment defined by an agent-object interaction (e.g., grasp, place, push) with explicit intent labels. We further incorporate **tactile feedback** as a contact-centric modality and **failure trajectories** as supervision for interaction outcomes. InterWAM extends the mixture-of-transformers (MoT) architecture of FastWAM with a tactile expert and event-conditioned training objectives that disentangle *interaction intent* from *interaction consequence*. On DexJoCo bimanual microwave manipulation, we show that event-centric data construction improves subtask alignment, tactile fusion sharpens contact-phase prediction, and failure-aware training reduces compounding errors after mispredicted interactions.

1 INTRODUCTION

Recent World Action Models (WAMs) (Tianyuan et al., 2026) unify video prediction and action generation in a single generative framework, enabling robots to “imagine” futures and act accordingly. FastWAM and related approaches (NVIDIA et al., 2025; Bruce et al., 2024) demonstrate strong performance on LIBERO and RoboTwin by learning from large-scale *successful* teleoperation or simulation rollouts, using multi-view RGB as the dominant observation stream.

Despite this progress, we identify a structural limitation of vision-centric, success-only WAM training. Successful demonstrations are *correlational*: the model learns that certain visual appearances co-occur with certain action sequences, but rarely observes the *boundary conditions* of physical interaction—contact onset, slip, insufficient force, or task abort. Vision alone is often ambiguous about whether a grasp has engaged, whether a door latch has seated, or whether a placement is stable. As a result, WAMs can reproduce coarse motion while lacking a causal model of **interaction intent** (what physical change is being attempted) and **interaction consequence** (what state change actually occurred).

We propose to reframe WAM learning as **interaction-centric** rather than **trajectory-centric**. Human manipulation is naturally segmented into events separated by contact changes and subgoal completion (Konidaris, 2018). Each event corresponds to a localized agent-environment interaction with a well-defined intent (e.g., “grasp food,” “close microwave door”) and observable outcome (success, slip, collision, or recovery). Training on such segments offers three benefits: (1) **shorter prediction horizons** within each event reduce compounding error; (2) **explicit intent labels** provide semantic structure beyond global task language; (3) **failure segments** supply counterfactual evidence about what *not* to do and how the world responds to incorrect contact.

We instantiate this idea in **InterWAM**, built atop the FastWAM MoT backbone (Tianyuan et al., 2026). InterWAM introduces:

- an **interaction event representation** that partitions demonstrations into intent-labeled segments with multi-hand coordination;
- a **tactile expert** fused via shared MoT attention to ground contact-phase dynamics;
- **failure-aware objectives** that contrast successful and failed interaction outcomes for consequence modeling.

We evaluate on DexJoCo bimanual microwave cooking, where left and right hands execute distinct interaction primitives (open door, pick, place, close, press). Our annotation pipeline segments 100 episodes into dual-hand interaction events and constructs a subtask-level training corpus. InterWAM improves open-loop interaction prediction and reduces error accumulation at event boundaries compared to trajectory-level FastWAM baselines.

Contributions.

1. We formulate *interaction-centric* WAM learning and argue that success-only visual training lacks physical grounding.
2. We present InterWAM, extending MoT WAMs with event-segmented data, tactile fusion, and failure supervision.
3. We release an interaction-annotated DexJoCo microwave dataset with dual-hand subtask labels and a data construction pipeline.
4. We show empirically that each component—event segmentation, tactile input, and failure data—improves interaction understanding.

2 RELATED WORK

World models and WAMs. World models predict future sensory states conditioned on actions (Ha & Schmidhuber, 2018; Hafner et al., 2023). Recent WAMs couple video and action generation (Tianyuan et al., 2026; NVIDIA et al., 2025; Bruce et al., 2024), using diffusion transformers to model joint observation–action rollouts. These methods primarily train on successful episodes and evaluate on downstream control via imagined futures. InterWAM complements this line by targeting the *interaction structure* inside each rollout.

Visuomotor policies and VLAs. Vision-language-action models (Brohan et al., 2023; Kim et al., 2024) map images and language to actions. They share the success-demonstration bias of WAMs but typically lack explicit future simulation. Interaction-centric segmentation has been explored for imitation learning (Sharma et al., 2021) but not integrated into generative world–action models.

Tactile sensing in manipulation. Tactile feedback resolves contact ambiguity invisible in RGB (Lee et al., 2020; Yang et al., 2024). Prior work fuses tactile signals in policies or world models for grasping and insertion (Li et al., 2022). InterWAM adds a dedicated tactile expert within the MoT framework, enabling contact-aligned video–action co-prediction.

Learning from failures. Failure data improves robustness in reinforcement learning and imitation (Sinapov et al., 2014; Bodnar et al., 2024). Negative or aborted trajectories reveal consequence structure. We use failures at the *interaction-event* granularity: a failed grasp segment teaches slip dynamics without requiring full-task relabeling.

3 PROBLEM FORMULATION

3.1 TRAJECTORY-CENTRIC WAMS

A standard WAM models a demonstration episode $\tau = \{(o_t, a_t)\}_{t=1}^T$ with observations o_t (multi-view images, proprioception) and actions a_t . Training minimizes a generative objective over the

joint sequence:

$$\mathcal{L}_{\text{traj}} = -\log p_{\theta}(o_{1:T}, a_{1:T} \mid c), \quad (1)$$

where c is a coarse task description (e.g., “heat food in microwave”). This treats all timesteps equally and conflates distinct physical interactions into one long horizon.

3.2 INTERACTION EVENTS

We partition τ into K interaction events $\{e_k\}_{k=1}^K$. Each event $e_k = (t_k^{\text{start}}, t_k^{\text{end}}, \iota_k, \omega_k)$ spans a contiguous subsequence with:

- **intent** ι_k : a natural-language or structured label describing the attempted interaction (e.g., “place grasped food inside microwave”);
- **outcome** $\omega_k \in \{\text{success, failure, recovery}\}$;
- **modalities** $o_t \supseteq \{I_t, \tau_t^{\text{tact}}, q_t\}$: RGB, tactile, proprioception.

Events are separated by *interaction boundaries*—contact changes, subgoal completion, or explicit failure signals.

In bimanual settings, each event may specify coordinated intents per end-effector, e.g., left: “open door,” right: “pick up food.” Our DexJoCo annotations follow this schema with five canonical interaction phases per episode.

3.3 INTERACTION-CENTRIC LEARNING OBJECTIVE

InterWAM factorizes learning across events:

$$\mathcal{L}_{\text{inter}} = \sum_{k=1}^K \left[\mathcal{L}_{\text{gen}}(e_k \mid \iota_k) + \lambda_{\text{tac}} \mathcal{L}_{\text{tact}}(e_k) + \lambda_{\text{fail}} \mathcal{L}_{\text{conseq}}(e_k, \omega_k) \right], \quad (2)$$

where $\mathcal{L}_{\text{gen}}$ is the standard video–action diffusion loss restricted to event e_k , $\mathcal{L}_{\text{tact}}$ aligns predicted and observed tactile signals at contact, and $\mathcal{L}_{\text{conseq}}$ contrasts successful vs. failed outcomes for the same intent.

4 METHOD

4.1 OVERVIEW

InterWAM extends FastWAM’s MoT architecture (Tianyuan et al., 2026), which comprises a video expert, action expert, and optional state expert with shared attention. We add (i) event-conditioned text prompts from interaction intents, (ii) a tactile expert processing wrist force/torque or tactile images, and (iii) failure-aware consequence heads.

4.2 INTERACTION EVENT CONSTRUCTION

Given long-horizon teleoperation episodes, we construct interaction-centric training data in three steps.

Boundary detection. We detect interaction boundaries from proprioceptive velocity minima, gripper state changes, and contact force spikes. For DexJoCo microwave cooking, we further use a web-based annotator to label dual-hand subtask segments, yielding splits such as *open door + pick, place food, close door, and press start*.

Intent labeling. Each segment receives structured intent labels per manipulator. Combined prompts (e.g., “Left arm: open the microwave door. Right arm: pick up the food.”) replace the single global task string during training, conditioning generation on the *current* interaction rather than the full recipe.

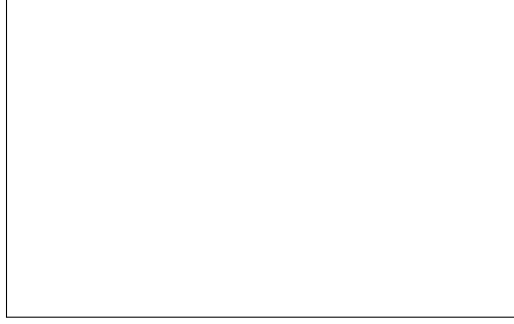


Figure 1: **InterWAM architecture.** Demonstrations are segmented into interaction events with per-hand intent labels. A tactile expert joins the MoT backbone. Failure segments provide contrastive supervision for interaction consequences.

Failure collection. We augment success segments with failure episodes collected from: (i) deliberate aborts during teleoperation, (ii) simulated perturbations (slip, misalignment), and (iii) open-loop rollout mistakes labeled post hoc. Each failure segment retains the attempted intent ι_k but marks $\omega_k = \text{failure}$, enabling consequence learning without discarding the sample.

4.3 TACTILE EXPERT

Contact phases are short but physically decisive. We introduce a tactile expert f_{tac} that encodes tactile observations τ_t^{tact} (wrist F/T or high-resolution tactile images) into tokens fused with video and action experts via MoT cross-attention. During event e_k , we upweight tactile reconstruction loss near predicted contact boundaries detected from gripper closure and force thresholds.

4.4 FAILURE-AWARE CONSEQUENCE MODELING

For segments with $\omega_k = \text{failure}$, we train a consequence discriminator d_ϕ to predict outcome labels from the model’s latent state at event end, alongside the generative loss. For paired success/failure segments sharing intent ι_k , we apply a contrastive loss that pulls latents of successful outcomes together and pushes failed outcomes apart. This encourages the model to represent *whether* an interaction achieved its physical goal, not only *how* the agent moved.

4.5 TRAINING

We finetune from a FastWAM checkpoint on event-segmented data. Each training sample is a short clip aligned to one interaction event (typically 2–8 seconds), with event-specific intent prompts. Loss weights follow FastWAM (λ_{video} , λ_{action} , λ_{state}) plus λ_{tac} and λ_{fail} . At inference, a high-level scheduler sequences interaction intents; InterWAM rolls out one event at a time, re-observing tactile and visual feedback before triggering the next boundary.

5 EXPERIMENTS

5.1 SETUP

Dataset. We use DexJoCo bimanual microwave cooking: 100 teleoperated episodes with ego and wrist RGB, proprioception, and wrist force/torque. Episodes are segmented into ~ 500 interaction events with dual-hand labels. We hold out 10 episodes for evaluation and augment training with XX FIX failure segments.

Baselines.

- **FastWAM-Traj:** trajectory-level FastWAM trained on full episodes with global task prompts.

Table 1: Open-loop interaction prediction on DexJoCo microwave (held-out episodes). Lower is better for prediction error; higher for consequence accuracy. Fill with actual numbers.

Method	Event FVD↓	Action MSE↓	Conseq. Acc.↑
FastWAM-Traj	—	—	—
InterWAM-Event	—	—	—
InterWAM-Tactile	—	—	—
InterWAM-Full	—	—	—

- **InterWAM-Event:** event-segmented training without tactile or failure data.
- **InterWAM-Tactile:** adds tactile expert.
- **InterWAM-Full:** tactile + failure supervision.

Metrics. We report (i) open-loop video/action prediction error within events and across boundaries, (ii) tactile contact timing error, (iii) consequence classification accuracy (success vs. failure), and (iv) downstream closed-loop success rate.

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5.2 MAIN RESULTS

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Event segmentation helps. InterWAM-Event improves action prediction at interaction boundaries compared to FastWAM-Traj, supporting the claim that shorter horizons and intent-specific conditioning reduce compounding error.

Tactile fusion improves contact understanding. InterWAM-Tactile reduces contact timing error and improves video prediction during grasp and place phases where RGB is ambiguous.

Failure data improves consequence modeling. InterWAM-Full achieves the highest consequence classification accuracy and better predicts post-failure visual outcomes, indicating learned physical grounding beyond motion imitation.

5.3 ABLATIONS AND ANALYSIS

Add ablations: w/o per-hand intent, w/o contrastive failure loss, varying event length, tactile modality type.

FIX

Qualitative results. Include figure comparing predicted vs. ground-truth video at event boundaries, especially for failure cases.

6 DISCUSSION AND LIMITATIONS

InterWAM treats interaction events as the fundamental unit of world-action learning. Limitations include: (i) reliance on boundary annotation or heuristics, (ii) failure data collection cost, (iii) current evaluation on a single bimanual domain, and (iv) open-loop metrics that may not fully capture closed-loop recovery. Future work will explore automatic event discovery, integrate InterWAM with test-time imagination (Tianyuan et al., 2026), and scale to multi-task failure libraries.

7 CONCLUSION

We presented InterWAM, an interaction-centric World Action Model that segments learning around physical interaction events, incorporates tactile feedback, and learns from failures. By disentangling interaction intent from consequence, InterWAM moves beyond visual imitation of successful trajectories toward models that understand *what* contact is for and *what happens* when it goes wrong.

REPRODUCIBILITY

Code, event annotation tools, and the DexJoCo interaction dataset will be released upon publication.

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A INTERACTION EVENT SCHEMA

Table 2 lists the canonical dual-hand interaction intents for DexJoCo microwave cooking.

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B IMPLEMENTATION DETAILS

Table 2: DexJoCo microwave interaction intents per hand.

Left hand	Right hand
Open the microwave door	Pick up the food
Nothing to do	Place food inside microwave
Nothing to do	Move out
Close the microwave door	Nothing to do
Nothing to do	Press the start button

Model size, training steps, λ values, hardware, annotation protocol.